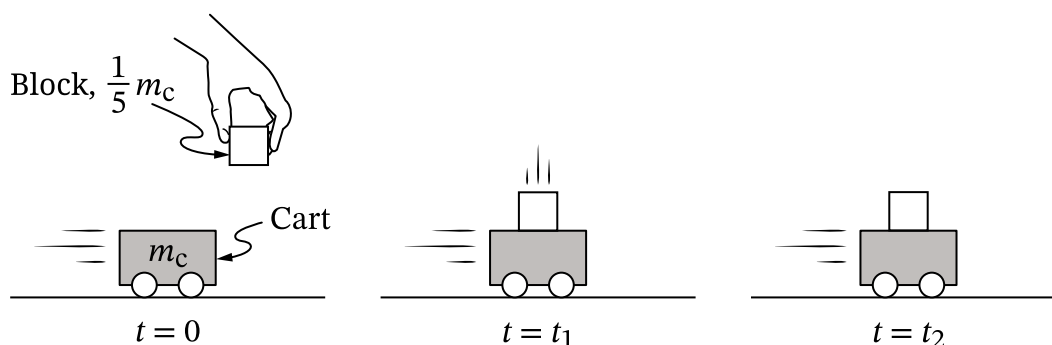


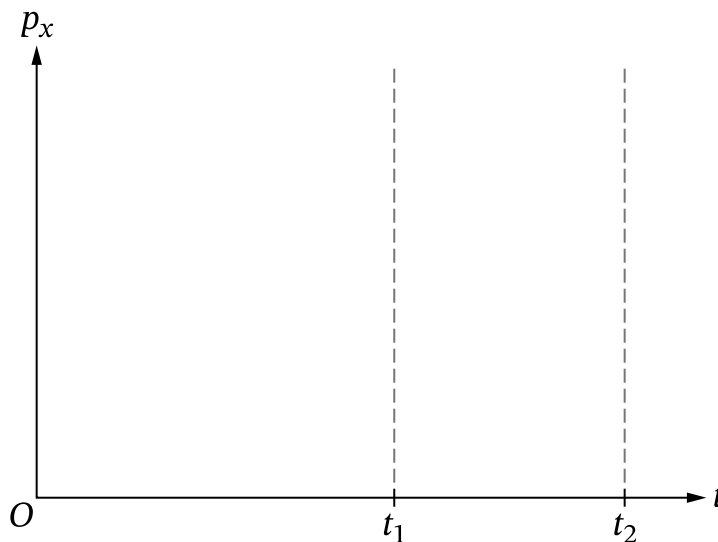
Question 1: Version J

1. A student has a cart of mass m_c and a block of mass $\frac{1}{5}m_c$, as shown in Figure 1.

- At time $t = 0$, the cart is moving to the right across a horizontal surface with constant speed v_c , and the student releases the block from rest.
- At $t = t_1$, the block collides with and sticks to the top of the cart. The block does not slide on the cart.
- At $t = t_2$, the block-cart system continues to move to the right with constant speed v_f .

Figure 1**A.**

- i. On the axes shown in Figure 2, **sketch** a graph of the magnitude p_x of the x -component of the momentum of the block-cart system as a function of time t from $t = 0$ until $t > t_2$.

**Figure 2**

- ii. **Derive** an expression for the speed v_f of the block-cart system after time $t = t_2$ in terms of m_c , v_c , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.
- iii. **Derive** an expression for the change in the kinetic energy ΔK in the block-cart system from $t = 0$ to $t = t_2$ in terms of m_c , v_c , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.
- B.** Consider the case where a new block is dropped and collides with the top of the cart. The new block slides along the cart during the collision but does not slide off the cart. The time interval from when the new block collides with the cart and moves together with the cart is Δt . During Δt there is a frictional force between the new block and the cart.
- Indicate** whether the x -component of the momentum of the new block-cart system increases, decreases, or remains constant during Δt .
- _____ Increases
_____ Decreases
_____ Remains constant
- Justify** your response.